

descriptions. This capability allows businesses to create customised visuals for advertisements, websites, and social media posts without requiring expensive design software or skills in graphics design. Companies can input specific requests—such as ‘a futuristic city skyline at sunset’ or ‘an abstract image of a digital cloud’—and receive high-quality images generated by AI. These tools are enabling companies to quickly create high-quality, eye-catching visual content for their marketing efforts.

The impact of open source AI goes beyond just text and images—it is also being used in video generation. By combining LLMs and VLMs, businesses can create videos from written scripts, revolutionising industries like entertainment and education. For example, open source models can take a script, automatically generate a storyboard, and then create animations or video sequences to match. This can significantly reduce the cost and time involved in video production while making it easier to create custom content at scale.

Customer service: Open source generative AI models are playing a pivotal role in enhancing customer service experiences. AI chatbots and virtual assistants powered by models like GPT-3 or specialised domain models can handle a wide range of customer queries, reducing the burden on human agents and enabling companies to provide 24/7 customer support. These models can understand natural language and generate human-like responses, making interactions smoother and more efficient.

Beyond basic query handling, generative AI is being used for more complex tasks, such as sentiment analysis and personalised

recommendations. For example, a customer might reach out to a support agent with a technical problem. An AI-powered system can analyse past interactions, determine the customer’s mood, and offer a personalised solution. Open source models such as RAG agents are particularly useful in these situations, as they combine real-time information retrieval with natural language generation, ensuring that responses are accurate and contextually relevant.

Moreover, voice interfaces are becoming increasingly popular in customer service, and open source speech-to-text and text-to-speech models are enabling seamless communication. Customers can interact with AI-driven systems through voice, making it easier to solve problems hands-free. SLMs help these systems transcribe spoken words into text and vice versa, making voice-based customer service both effective and efficient.

Retail and e-commerce: The retail industry is also benefiting from open source generative AI through personalised shopping experiences and automated inventory management. Open source AI models are being used to create recommendation systems that suggest products to customers based on their browsing history, preferences, and even social media activity. These recommendation engines drive customer engagement and sales by presenting highly relevant products to shoppers, boosting conversion rates.

Furthermore, generative AI is enhancing the customer review analysis process. AI models can process and generate summaries of customer feedback, identifying trends, sentiments, and potential issues, all of which can inform product development and marketing strategies. In e-commerce, visual search engines powered by generative

AI enable customers to upload images and receive similar product recommendations, streamlining the online shopping experience and enhancing customer satisfaction.

The challenges of open source generative AI

The open source movement in generative AI has undeniably democratised access to cutting-edge technology. However, with its transformative potential come significant challenges that require careful attention and actionable solutions.

Bias and fairness in open source generative AI: Generative AI models rely heavily on the data used for their training. In the case of open source systems, these datasets are often curated from diverse, publicly available sources, including text, images, and other multimedia content. Unfortunately, the biases embedded in these sources can inadvertently seep into the models, resulting in outputs that perpetuate stereotypes or exclude underrepresented perspectives.

For example, a generative AI system trained on internet text may disproportionately represent dominant cultural narratives while neglecting voices from marginalised communities. Such biases not only undermine the fairness of these models but can also lead to reputational risks for organisations adopting them.

Potential solutions are:

- Implementing bias detection frameworks that continuously analyse and flag skewed model behaviour.
- Encouraging community-driven curation of training datasets to ensure they are representative and diverse.
- Leveraging fine-tuning techniques to adapt general models to specific use cases with fairness considerations in mind.